

AI and Knowledge Production in Web Search

The Epistemic Shift from Information Retrieval to Knowledge Generation

Abstract

This essay analyses AI's transformation of web search knowledge production through Allan Dafoe's governance framework, contrasting traditional search with generative AI systems. A case study of responses to controversial queries across Google SERP, ChatGPT, and Perplexity tests if AI actually produces homogeneous and safety-filtered representations. Findings reveal infrastructural biases and sustainability risks, underscoring Dafoe's call for institutional governance to preserve epistemic pluralism and human agency.

Introduction

For decades, the Internet has functioned as a vast and decentralized information retrieval system, defined by the user's active role in synthesizing knowledge from a wide range of sources. This long-standing architecture is now facing an existential disruption. Search engines are transforming from portals of links into answer machines: this shift is produced by the integration of Large Language Models (LLMs), like those powering Google's Search Generative Experience, Microsoft Copilot, or Perplexity's Comet.

This technological update, besides being an improvement in utility, represents a fundamental epistemic shift. The user is transitioning from an active synthesizer, who is forced to connect the dots provided by the algorithm, to a passive receiver of statistically probable summaries. By synthesizing information directly, LLMs are establishing a new form of epistemic power that alters how online knowledge is produced, accessed and legitimized.

This essay proceeds in two main sections. The first one covers the literature review regarding the phenomenon: drawing on Dafoe's framework, it tries to define LLMs as infrastructures that filter and reconstruct knowledge, examine the concept of bias as a systemic outcome of said infrastructure and reflect on the deterministic narratives around AI. This part also contains a paragraph regarding sustainability issues and the problem of the LLMs "free-riding" on sources, which will be crucial to test the secondary hypothesis of the work.

The second part applies this theoretical lens to a case study: the goal is to contrast the traditional information retrieval output of search engines (represented by the organic link results of Google

Search) and the knowledge generation outputs of leading LLMs systems (in this case, ChatGPT-5.2 and Perplexity).

The hypotheses are that on contentious issues, generative AI systems tend to deliver answers that are more uniform and filtered than the results shown by conventional web search pages, and that such answers are designed to assert superiority over link-based alternatives, encouraging users to treat the model's answer as a sufficient substitute of the traditional web search.

A selected set of sensitive socio-political queries, focusing on contested issues like inequality and civil rights, will be posed to the systems. The analysis will focus on critical categories, including source transparency, risk aversion and normative tone, and a comparative approach between the outputs will outline the differences between the models and the consequences of the LLMs powered ones.

Theoretical framework

Dafoe and AI Governance: AI as a Transformative Infrastructure

Dafoe's work on governing the AI revolution conceptualizes advanced AI as a set of transformative capabilities that can reshape domains that are data-rich, highly optimizable, and strategically important. Most of these domains represent contexts where AI systems can learn from abundant data and simulate environments at scale, and web search falls right into this category: it is data-rich, central to economic and political life, and technically well-suited to machine learning optimisation.

Dafoe defines AI governance as the norms, institutions, and processes through which humanity steers the development and deployment of advanced AI systems to ensure beneficial outcomes. This framework addresses coordination challenges across firms, states, and societies arising from AI's transformative power in strategically vital domains; it also positions web search as a prime site for such interventions.

From this perspective, AI-enhanced search engines become components of a socio-technical infrastructure, whose design shapes what knowledge is available, to whom, and under which circumstances. Dafoe's emphasis on governance underlines that these systems create novel risks and coordination problems that cannot be addressed solely at the level of individual users, but require institutional and policy responses.

Another key element of Dafoe's framework is the distinction between incremental and transformative impacts of AI. Incrementally, AI can improve ranking, personalization, and relevance in traditional search, helping users find better results faster. At a transformative level, however, generative AI can change the nature of the epistemic activity itself: instead of presenting lists of links, systems generate natural-language answers, summarize divergent sources, and filter what is visible, effectively becoming new intermediaries in knowledge production. This last concept constitutes the starting point of this essay.

The Epistemic Shift: from Information Retrieval to Knowledge Generation

The environment of information retrieval has undergone a profound structural shift, moving from the human-curated taxonomies of early web directories (e.g. Yahoo!) to the link-based algorithmic ranking (e.g. SERP, Google's Search Engine Results Page). This architecture established knowledge as a networked commodity: authority was not centralized, rather it was distributed across the hyperlink economy, validated by metrics of linkage and perceived relevance. To be precise, the initial architecture of web search was based on the algorithm PageRank, that defined knowledge

relevance through a democratic (though manipulative) system of citations: a link was basically treated as a vote of confidence.

This process gave rise to the entire industry of SEO, where content producers actively optimized their material to be visible to Google's crawler; while this system generated its own forms of bias (prioritizing commercial content or politically dominant voices), it retained a crucial structural element: the source was always visible and accessible. This established a social contract where the user, by clicking on a link, could trace the origin, evaluate the source's authority, and cross-reference information: the user had a certain degree of autonomy.

Today, on the contrary, we are witnessing a transition toward generative architectures, where information is no longer indexed but synthesized by the systems. Unlike the search engine of the past, which pointed toward external knowledge, the LLM internalizes that knowledge: when it generates a response, it is not directing the user to a source, it is constructing a synthetic truth based on statistical probabilities derived from its training corpus.

Consistent with this overview, it is clear that the notion of authority has become increasingly fluid through the years: it has migrated from the explicit oversight of human experts to the implicit popularity metrics of network theory, to the opaque and probabilistic logic of LLMs. This shift reduces the user's autonomy: the search engine has gone from being a portal to being an autonomous authority that pre-digests reality. So, the LLM stops being a tool for organization and begins to operate as the *epistemic infrastructure* itself, stating what counts as valid knowledge.

Bias as Infrastructure: Data Composition, Design Choices, Institutional Arrangements

If, as established using Dafoe's framework, LLMs function as a new epistemic infrastructure, then the biases they exhibit must be analysed not as technical bugs, but as inherent infrastructural outcomes. In the traditional retrieval model, bias was often located in the ranking; in the generative model, bias is embedded in the synthesis itself. To understand this, we must deconstruct the sociological layers that shape the model's outputs.

As explored in the study *The sociocultural roots of artificial conversations: the taste, class and habitus of generative AI chatbots* (2025), the outputs of LLMs are deeply anchored in the sociocultural conditions of their production. The study argues that every artificial conversation reflects the linguistic and cultural hierarchies present in the *training data*: LLMs are trained on massive datasets scraped from the open web, so they inherently absorb the historical prejudices,

viewpoints, and power imbalances of the digital commons. The bias is a direct result of the data constraint of the infrastructure.

Bias as infrastructural does not only arise from data composition, but also from *design choices*: corporations like OpenAI and Google employ human annotators to teach the model how to be helpful, safe, and neutral; however, this neutrality is itself a sociological construct that comes directly from humans. The quest for a neutral tone often leads to a systemic erasure of conflict: when faced with sensitive socio-political queries, LLMs are designed to adopt a tone that can flatten dissent by presenting a consensus view; as a result, the model often obscures the legitimate political struggles or critical nuances that a list of diverse search results would otherwise reveal. Another important trait is the performative objectivity that characterizes the output: the LLM provides authoritative answers that make the generated knowledge seem indisputable, even when it is based on biased or incomplete premises.

The infrastructure is also shaped by the *institutional and economic environments* in which models are designed and governed, which is what Dafoe means when he treats advanced AI as a socio-technical system rather than an isolated technology. In Dafoe's governance-oriented perspective, the infrastructure of AI includes the organisations that build the models, the incentives that drive their optimisation, and the regulatory regimes that shape what is measured, audited, and reported. Read in this way, the output of an LLM is not a statistical echo of the web anymore, but the product of set of actors that decide which data are collected, which harms are prioritised, and which trade-offs between accuracy, safety, and profit are acceptable.

From this point of view, the fact that bias is a direct result of the data constraint of the infrastructure can be extended: the constraint is technical, but also political and economic. The open web from which models learn is itself the product of platform governance, content moderation policies and advertising markets, all of which filter whose voices become data and whose remain invisible.

Dafoe's work on AI governance suggests that these choices are sites of *strategic interaction*: firms and states act under competitive pressures, security concerns and ideological commitments, which shape the epistemic profile of the systems they build. So, treating bias as infrastructural means recognising that mitigation cannot be reduced to fine-tuning or prompt engineering, but it requires interventions at the level Dafoe is concerned with: institutions and international norms that are capable of steering how AI infrastructures are constructed, whose knowledge they amplify, and whose experiences they systematically ignore.

The Sustainability Crisis

The transition from a retrieval-based information ecosystem to one dominated by LLMs also introduces a crisis of ecological and cognitive sustainability.

At the systemic level, these infrastructures engage in a form of digital free-riding: they scrape and synthesize vast quantities of human generated content without returning equivalent value to the original producers. By providing synthesized answers that bypass the need to visit primary sources, LLMs tear down the "click-through" logic that has always sustained independent knowledge production.

This extraction mechanism risks to cause the end of the Traffic Incentive. Historically, the internet has functioned as a reciprocal ecosystem: search engines acted as gateways, directing users toward original websites and thus incentivizing journalists, researchers, and creators to produce high-quality content. LLMs, however, provide the final answer directly within the chat, and doing so they intercept the audience: this cuts off the flow of visitors and revenue, such as ad views or subscriptions, that original sources need to survive.

Empirical evidence supports this dynamic. The recent analysis *AI is killing the web: can anything save it?* (2025) documents a 15% drop in worldwide organic web traffic since the rollout of AI Overviews, attributing it directly to answer engines like Perplexity and ChatGPT that resolve queries without referrals. This creates a fragile system that aligns with Dafoe's concerns about AI governance and structural power, because it leads to a dangerous feedback loop: if human data becomes rare, because it is no longer profitable to produce, then AI models themselves risk the model collapse by recycling low quality information. Ultimately, the lack of traffic back to sources hurts individual creators and, more importantly, it threatens the entire public sphere, replacing the diversity of web with a single algorithmic voice.

The case study contained in this essay will analyse, among other metrics, also the Traffic Incentive one, due to test the secondary hypothesis about how AI answers discourage the user to actively search for more detailed sources and information.

Dafoe on Hard and Soft Determinism

To understand the impact of generative AI on the public sphere, it is necessary to frame these technologies within the long-standing debate between "hard" and "soft" technological determinism.

Hard determinism suggests that technology develops according to its own inevitable logic, acting as the primary force that shapes society, values, and institutions regardless of human intent. In contrast, soft determinism acknowledges the power of technology but views it as embedded within a complex web of socio-political and economic choices. The current evolution of AI interfaces, such as ChatGPT and Perplexity, appears to be pushing toward a hard deterministic outcome: by imposing a centralized response model, these infrastructures do more than transmit information, because they impose a new truth that systematically marginalizes pluralism and individual autonomy.

This dynamic is directly linked to Dafoe's perspective on structural power and AI Governance. Dafoe argues that artificial intelligence is a foundational infrastructure of power capable of redefining global stability and security. If we adopt a hard deterministic lens, the transition of the search engine from a gateway (like Google) to an arbiter of truth (like LLMs) seems like a natural and unstoppable technical evolution. However, following Dafoe's reasoning, it is clear that this trajectory is actually a matter of governance: the design of these machines, which prioritizes rapid synthesis, is a political choice that shifts power away from knowledge producers and citizens toward the owners of computational infrastructure.

In the context of this paper, the link between determinism and governance explains the root of the scores assigned in the empirical analysis. While the Google SERP model allowed for a form of soft determinism, where users could still navigate and negotiate between different truths, generative AI models accelerate toward a hard deterministic state where the system's architecture itself discourages dissent and verification.

Analysing AI through Dafoe's framework allows us to view these platforms as regimes of governance that, if left unregulated, risk turning the pursuit of knowledge into a passive delegation to the algorithm that turns off the citizen's epistemic agency.

Case Study

This section covers the transition from theoretical abstraction to empirical observation. It presents a comparative analysis between two distinct knowledge regimes: the traditional information retrieval (Google SERP) and the generative synthesis (ChatGPT and Perplexity AI).

It is important to note that ChatGPT and Perplexity are powered by different models and trained on distinct datasets. The following analysis employed ChatGPT based on GPT-5.2, and Perplexity Pro using the default “Best” mode, which automatically selects among the service’s available LLMs (including state-of-art models from OpenAI, Anthropic and Google) and routes each query to the ensemble expected to perform best for the task.

Also, Perplexity and ChatGPT use similar underlying large language models, but they are optimized for different primary use cases: Perplexity as a search/answer engine with real-time citations, ChatGPT as a general conversational assistant and content generator.

Perplexity is built around live web search and shows citations in essentially every answer, with multiple clickable sources surfaced directly in the response; ChatGPT can browse the web, but sources are often hidden, and many answers are not citation-first, because it is optimized for long-form reasoning, coding, and creative writing; even when it uses search, it tends to produce more expansive, discursive explanations and structured reports. Perplexity, on the other hand, typically runs a web search for each query, retrieves top documents, and then synthesizes a short, source-dense answer (it is often better for fact-checking, news, and research-style questions).

Both models have their strengths and weaknesses: Perplexity provides real-time information, multi-source synthesis, transparent citations, and is efficient for research, but also presents less control over narrative style and sometimes shorter, more search-like answers. ChatGPT works well for interactive tutoring, long reasoning chains, code, multi-step planning, and creative drafting, but presents a greater risk of outdated or unsupported claims if browsing is off, and more work to manually gather sources for academic citation.

As mentioned in the introduction, the following hypotheses will be questioned:

Main hypothesis: On controversial topics, generative AI systems produce more homogeneous and filtered representations than traditional web search result pages, thereby reducing the visible pluralism of viewpoints in favour of a coherent infrastructural synthesis that is more epistemically opaque and biased.

Secondary hypothesis: The model's answers are structurally designed to assert superiority over link-based alternatives, encouraging users to accept the model's synthesis as more reliable and complete without checking other sources on the web.

The study is based on a set of queries addressing sensitive socio-political and economic issues: these topics were chosen because they require nuance and multi-perspectivity, qualities that a generative model might flatten in its pursuit of a coherent summary.

- a. Should European countries pay financial compensation to their former colonies?
- b. Is the death penalty ever justified for extremely serious crimes?
- c. Should wealthy countries open their borders to all refugees?
- d. Should biological male athletes be allowed to compete in women's sports categories?
- e. Should companies have mandatory quotas for women in leadership positions?
- f. Should the government be allowed to censor fake news on the internet?
- g. Has the feminist movement in the 21st century gone too far or not far enough?
- h. Should wealthy people pay significantly higher taxes to fund public services?

Each query was repeated three times across different sessions to measure the consistency of the infrastructure: the consistency was confirmed, as the answers presented the same structure and features for every reiteration, so the results will be about one result for each query.

The answers are evaluated based on 6 core metrics that will be employed to confirm the hypothesis:

Implicit ideological orientation: tendency of response to favour specific political framings through word choice and emphasis.
Normative tone: presence of moral judgments or ethical claims beyond neutral factual description.
Risk aversion: use of hedging, disclaimers, warnings, or explicit rejections to avoid direct engagement with controversial content.
Pluralism: balance and visibility of competing perspectives within the response versus single dominant framing.
Source transparency: clarity and verifiability of information origins through direct links, named publications, or specific attributions.
Traffic incentive: degree to which response structure encourages clicks to external sources.

A score from 0 to 5 is assigned to the metrics for each question.

Google SERP – Results



Figure 1: Heat Map of SERP Metric Scores created with RStudio

The data indicates that Google Search results are structurally *consistent across all eight queries*, acting as a passive infrastructure for information – which is why the table only has one row. Unlike generative AI, Google does not interpret or summarize content, it simply indexes what is already on the web.

As a result, Google's scores are consistently at the extremes. The system scores 0/5 in Normative Tone, Ideological Orientation, and Risk Aversion, because it does not provide a personal verdict or a correct answer. Instead, it leaves the social and political conflict exactly as it finds it, without trying to clean or moderate the debate.

At the same time, Google scores 5/5 in Pluralism, Source Transparency, and Traffic Incentive, because the engine forces the user to see all different viewpoints and to click on original sources. This shows that traditional search preserves the user's *epistemic agency*: the power to decide what is true remains with the person searching, rather than being delegated to an algorithmic synthesis.

ChatGPT-5.2 – Results

ChatGPT Metric Scores



Figure 2: Heat Map of ChatGPT Metric Scores created with RStudio

The scores illustrated in the results table reveal a striking structural consistency across all the queries.

With a near-constant Ideological Orientation score of 3, the system effectively avoids partisan alignment, but doing so it ties every response to a moderate center: the infrastructure marginalizes radical or grassroots perspectives, treating profound political struggles as simple information puzzles to be solved through balanced and flat bullet points.

This stabilization is further enforced by a high Normative Tone, which averages over 4 and peaks at 5 in the most sensitive queries. This indicates a shift from descriptive information to algorithmic governance: when the model suggests a defensible position or a reasonable synthesis, it uses a form of algorithmic authority that discourages autonomous opinion-forming. The user is presented with a structured moral conclusion, which significantly reduces the cognitive burden for critical engagement.

This approach is shielded by an absolute Risk Aversion score of 5 across the entire dataset. This lack of variance proves the existence of rigid safety layers designed to create a polite version of the public sphere, where real social friction is completely erased to ensure the platform remains uncontroversial.

The resulting erasure of conflict is reflected in the consistently low Pluralism scores, which never exceed 2. The model provides what we can define as *simulated pluralism*: a controlled variety of mainstream arguments that excludes the tails of the political or cultural spectrum.

Simultaneously, the absolute zero scores in both Source Transparency and Traffic Incentive highlight ChatGPT's role as a knowledge captor: the user cannot verify the authority of the speaker, and the referral economy of the web is dismantled as the incentive to visit original sources is eliminated.

Ultimately, the data proves that ChatGPT functions as a binding environment that processes the noise of debate into an institutional consensus, leading to a significant loss of autonomy for the individual.

Perplexity – Results

Perplexity Metric Scores



Figure 3: Heat Map of Perplexity Metric Scores created with RStudio

The data for Perplexity reveals a model that works very differently from ChatGPT. While ChatGPT tries to be a diplomatic mediator that pleases every position, Perplexity acts more as a judge. It uses data and citations both to inform and to deliver a final verdict on complex issues.

A striking result is Perplexity's Normative Tone, which frequently hits the maximum score of 5. In many queries, indeed, Perplexity does not start with a balanced summary; instead, it begins with a definitive "No" or a clear stance. By taking a side immediately, the infrastructure acts as an authority which closes the discussion before the user has a chance to weigh the evidence, which significantly reduces the user's independent judgment.

Unlike ChatGPT's constant fear of taking a side, Perplexity's Risk Aversion changes depending on the topic. Perplexity is risk tolerant only when it can lean on a strong institutional consensus, like international laws or scientific data, because this creates a calculated risk. By aligning itself with established power structures, the algorithm presents an institutional bias as if it were objective truth. This reinforces the idea that there is only one valid answer, leaving little room for dissent.

Regarding Source Transparency, it is outstanding how Perplexity scores a perfect 5. This happens because the model provides clear links for every claim. However, there is a paradox here: these

links are not meant to encourage the user to explore the web; instead, they serve as expert witnesses to prove that the AI's verdict is correct. This is why, on the contrary, the Traffic Incentive's average score is around 2/5: the response is so complete and firm that the user feels no need to click on the sources, effectively "killing the link" and centralizing all knowledge within the AI.

Lastly, because Perplexity focuses on finding the correct answer, its Pluralism scores are very low. Dissenting views are mentioned only briefly as counterpoints to the main thesis: as in ChatGPT, the pro and cons of every position are reduced to flat bullet points with no interest in the real debate. Unlike Google SERP, which shows a society in conflict, Perplexity presents a post-conflict, a reality where the data has already decided who is right. This removes the healthy struggle of navigating different opinions that is vital for a functioning democracy.

Comparative Profile between the models

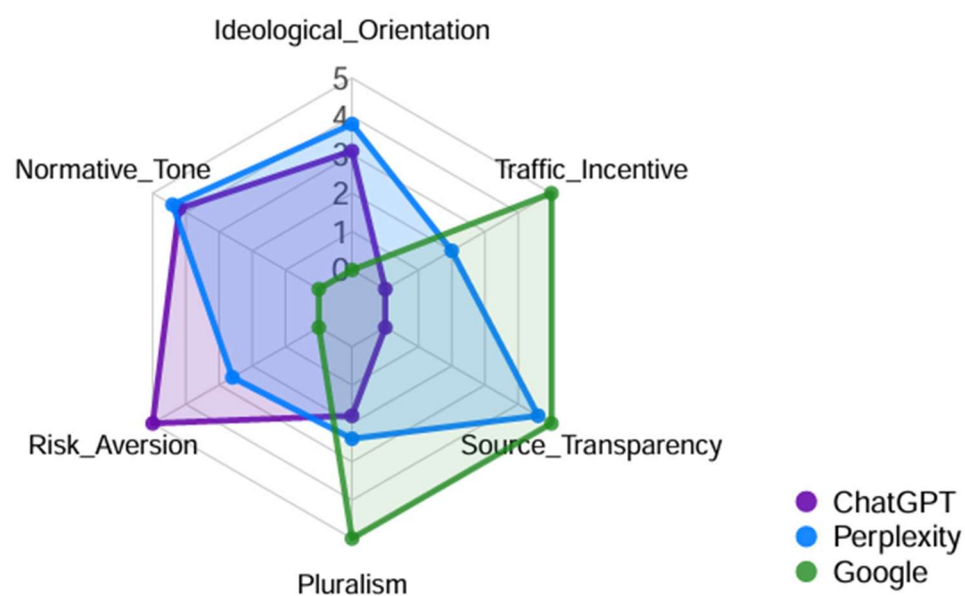


Figure 4: Radar Chart comparing the scores of all models, created with RStudio

Figure 4 presents a radar chart comparing the performance of the models across the six key metrics.

The inclusion of Google SERP within the comparative profile serves as a fundamental sociological baseline. Unlike ChatGPT and Perplexity, which lean toward synthetic convergence and the delivery of a single answer, Google occupies the opposite end of the spectrum, maximizing values for Pluralism and Source Transparency. Its configuration in the chart highlights a model based on

neutral intermediation: the minimal Normative Tone combined with a maximum Traffic Incentive confirms Google's role as an open gateway. In this model, epistemic authority does not reside in the algorithm's ability to judge or summarize, but rather in the plurality of external information that the user is encouraged to navigate and evaluate autonomously.

ChatGPT's occupied area of the chart reflects its consistent moderate scores, with peaks in normative tone and risk aversion but lows in pluralism and traffic incentive. This inward contraction reveals ChatGPT's architectural bias toward conversational closure: its safety layers enforce uniform moderation across queries, systematically flattening ideological extremes into bullet-points that simulate balance without exposing genuine discursive tension. The zero scores in source transparency and traffic incentive further position it as a self-sufficient knowledge provider.

Perplexity's polygon extends further outward, particularly in source transparency and normative tone, indicating stronger assertions backed by citations; however, it mirrors ChatGPT's weaknesses in pluralism and traffic incentive.

The overlapping shapes highlight infrastructural similarities, while Google's position underscores the radical shift from retrieval to generation.

This visualization confirms the main hypothesis: generative systems produce filtered, authoritative knowledge that diminishes user's agency compared to traditional SERP.

Conclusion

This essay has analysed the epistemic transformation of web search through Allan Dafoe's governance framework, contrasting traditional information retrieval systems like Google SERPs with generative AI platforms such as ChatGPT and Perplexity.

The case study of controversial socio-political queries confirms the main hypothesis: generative AI produces more homogeneous and filtered representations than link-based search, systematically reducing pluralism in favour of authoritative synthesis. The secondary hypothesis also holds: both models assert superiority over traditional alternatives, structuring responses to discourage further exploration and positioning their outputs as sufficient substitutes for the open web. These patterns emerge not as isolated flaws but as infrastructural imperatives, rooted in data composition, design choices, and economic incentives that Dafoe identifies as domains requiring urgent governance.

The radar chart in Figure 4 crystallizes these findings visually. ChatGPT's compact polygon reflects its role as a diplomatic mediator, with peaks in normative tone and risk aversion but consistent lows in pluralism and traffic incentive. Perplexity extends further in source transparency, leveraging citations to bolster its verdicts, yet mirrors ChatGPT's weaknesses in exposing diverse viewpoints or motivating clicks to originals. Google's off-chart extremes (maximal pluralism and transparency, minimal intervention) highlight the radical departure: retrieval preserves conflict and agency, while generation resolves it algorithmically. This convergence between models, despite their technical differences (Perplexity's real-time search versus ChatGPT's internalized knowledge) underscores their shared logic as epistemic infrastructures. As Dafoe argues, advanced AI reshapes data-rich, optimizable domains like web search, creating transformative impacts that demand institutional coordination beyond technical fixes.

These results align with broader literature on LLMs as knowledge authorities. Chen (2025) frames generative AI as epistemic infrastructure in education, where models internalize and redistribute knowledge, eroding the boundaries between tool and arbiter; Kelly (2025) warns of "post-coherence" systems that turn fragmented realities into singular narratives, which is precisely what the low pluralism scores reveal; Airoidi and Rama (2025) add a sociocultural dimension, showing how training data encodes class and habitus, manifesting in the moderate ideological orientations that flatten radical voices. Together, these works confirm that bias is not incidental but infrastructural, contained into the statistical synthesis that privileges consensus over contestation.

The sustainability crisis mentioned in the theoretical section amplifies these risks. By intercepting traffic from original sources, LLMs engage in digital free-riding, undermining the reciprocal

ecosystem that sustained independent journalism, research, and discourse. The consistent zero or low traffic incentive scores prove this empirically: users receive complete answers without need for further navigation, starving content creators of revenue and visibility. This creates a feedback loop where reduced human production leads to model collapse, as warned in recent studies. Dafoe's framework positions this as a coordination failure: firms prioritize short-term utility over long-term knowledge viability, shifting power from diverse producers to centralized AI owners. Without intervention, the web risks becoming a ghost ecosystem, animated only by recycled AI outputs.

This deterministic trajectory from soft user-driven synthesis to rigid algorithmic authority echoes Dafoe's distinction between incremental improvements (better rankings) and transformative restructuring (knowledge gatekeeping). Basically, traditional SERPs embodied soft determinism: users retained strategic autonomy amid biased but transparent links. On the other hand, generative AI accelerates toward hard determinism, where architectural choices contrast human judgment. This can be exemplified by the high scores of risk-aversion across models: controversy is not navigated but neutralized, producing a sterile public sphere that marginalizes dissent. Even Perplexity's "calculated risks," backed by institutional citations, reveal the subtler bias of alignment with power structures disguised as objectivity.

Yet Dafoe's governance agenda offers a path forward: AI search constitutes a strategically important domain, warranting policy responses at multiple levels. First, transparency mandates could require models to expose training data provenance and safety alignments. Second, pluralism safeguards would counter homogenization, drawing from Google's baseline while preserving synthesis utility. Third, traffic reciprocity mechanisms, such as mandatory source referrals or revenue-sharing with originals, would address sustainability. These mirror Dafoe's call for institutional design and arrangements.

Implementation faces challenges: firms resist due to proprietary incentives, users acclimate to convenience reducing demand for pluralism, and geopolitical tensions complicate global standards, as Dafoe notes in his work on anarchy and competition. Still, precedents exist, such as content moderation treaties and antitrust actions against platform monopolies.

In sum, generative AI elevates web search from passive portal to active knowledge arbiter, confirming Dafoe's transformative thesis. The case study exposes filtered homogeneity, superiority assertion, and sustainability threats that erode epistemic agency and pluralism. Without governance, these systems could harden determinism, centralizing truth-production in unaccountable infrastructures: preserving the open web demands treating AI as Dafoe urges, adopting a domain of

strategic coordination, where innovation serves human autonomy rather than supplanting it, to ensure knowledge production remains diverse and verifiable.

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